KAPASA MAKASA UNIVERSITY

DEPARTMENT OF INFORMATION COMMUNICATION TECHNOLOGY

CYS 121 INFORMATION SYSTEMS SECURITY

Duration 3 Hours

INSTRUCTIONS

1. Answer ANY 5 Questions
2. Any reference material is not permitted

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Question One

Explain along with examples the following Cyber Security terminologies which are termed the tenets of security

a) Confidentiality [5 Marks]
b) Integrity [5 Marks]
c) Availability of data. [5 Marks]
d) Non-Repudiation [5 Marks]

Question Two

Database Security is one of the core objective of Information Systems Security. Security of the database provides an assurance of confidence by data subjects. As a Cyber Security Specialist you have been called upon by the Born Before Computer Institute (BBIC) to explain the following database security requirements;

a) Physical Database Integrity [4 Marks]
b) Auditability [4 Marks]
c) Access control [4 Marks]
d) User authentication [4 Marks]
e) Availability [4 Marks]

Question Three

a) List three(3) authentication questions (but not the answers) a credit card company or bank could ask to authenticate a customer over the phone. The questions should be ones to which an impostor could not readily obtain the answers. [2 Marks Each]
b) For each of the questions state the reason why such a question would be important [2 Marks Each]
c) Identify and explain Four(4) examples of Biometrics authentication methods and state circumstance in which the use of such is very appropriate [2 Marks Each]

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Question Four

a) State and explain three(3) types of password attacks? What can a systems administrator do to protect against them? [6 Marks]

b) What are five(5) ways do administration or management play a role in access control of an Information System? [10 Marks]

c) State two (2) physical access controls [2 marks each]

Question Five

At times attack may happen without the organization realizing the extent of the damage. You are required to explain with examples the circumstance in which a real damage and high cost to the Information System(IS) may be regarded to have occurred:

Explain along these lines of reasoning

a) Criticality of information attacked [5 marks]
b) Availability of the IS [5 Marks]
c) Possibility of recovery [5 Marks]
d) Cost to the organization [5 Marks]

Question Six

Explain at least three kinds of harm that could be experienced from unauthorized modification of :

a) Medical information [5 marks]
b) Election information [5 marks]
c) Examination body data [5 marks]
d) Online order information [5 marks]

Question Five

Briefly discuss Ten (10) best practices of access control. [2 Marks Each]

ALL THE VERY BEST!